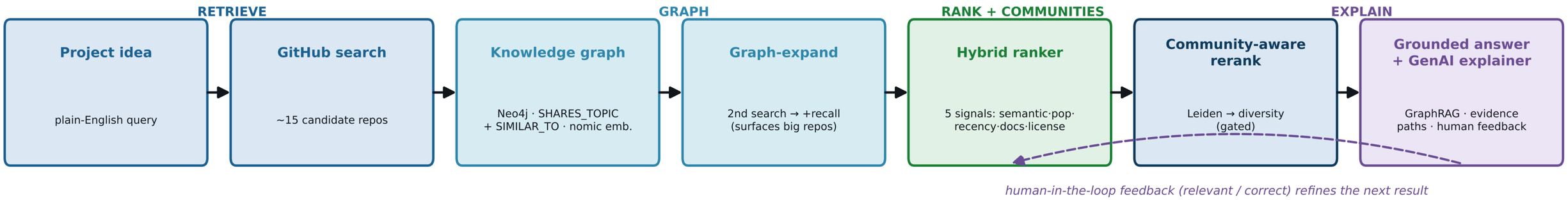
# RepoReason

## Graph-Augmented, Community-Aware, Explainable GitHub Repository Search

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Scan — project repo

## System pipeline — from a plain-English query to a grounded, community-aware answer




## Abstract

RepoReason ranks GitHub repositories by how well they fit a plain-English project idea — not by stars alone — and explains every result so users can trust it. For each query it builds a live Neo4j knowledge graph of candidate repos, uses the graph to recall more good repos, fuses five signals into a hybrid relevance score, detects communities with Leiden (Louvain fallback), and reranks for diversity — then answers follow-up questions with GraphRAG grounded in graph facts. On a hand-labeled benchmark the hybrid ranker reaches nDCG@10 0.78 (vs 0.72–0.74 for single-signal baselines), and community-aware retrieval raises result diversity +35% with no loss of relevance — turning a star-sorted, near-duplicate list into a varied, explainable, trustworthy shortlist.

## Key contributions

- A hybrid relevance ranker that fuses 5 signals and ranks best-or-tied on every metric (relevance margins within n=6 CIs).
- A per-search knowledge graph that drives recall — surfacing strong repos the keyword search misses.
- Community-aware reranking that lifts result diversity +35% (95% CI excludes zero) at no relevance cost.
- An evidence-driven Human–AI app: grounded GraphRAG answers + human-in-the-loop feedback.
- Honest evaluation with 95% CIs (n=6) — a robust diversity gain, directional relevance gains, and dead ends reported.

## Introduction

**The problem.** GitHub search ranks mostly by stars and literal keyword matches. Three failures follow: small-but-excellent repos get buried; repositories whose names merely echo the query win on “semantic echo” (a 5★ clone can outrank a 38k★ framework); and results fill with near-duplicates. The user never sees why a repo ranks or how the options relate.

**Target users.**

Students planning capstones, researchers scanning prior code, and developers choosing between libraries — anyone who needs the best-fitting repos for an idea, with a reason they can trust.

**Objective.**

Rank by genuine fit; group results into meaningful communities so the shortlist is diverse; and make every answer explainable and open to human correction.

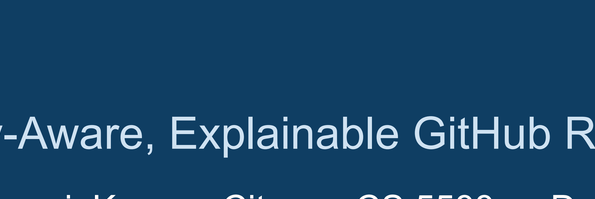
## Human–AI application




*Live app: community-grouped results, “why this fits”, and relevant / not / wrong-community feedback.*

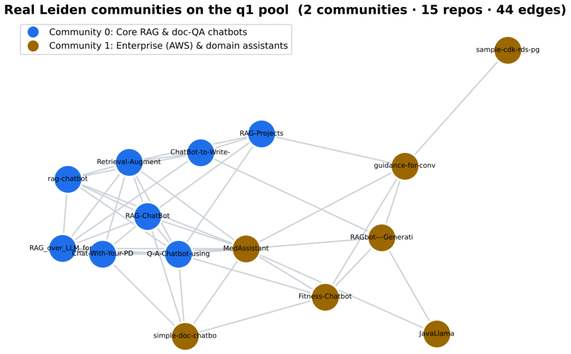
## Methodology

**Each search runs the pipeline above.** (1) A GitHub search returns ~15 candidates. (2) A Neo4j knowledge graph links them by shared topics (SHARES_TOPIC) and embedding similarity (SIMILAR_TO: cosine ≥ 0.60, top-5 neighbours) using nomic-embed-text. (3) The graph’s anchors and themes drive a second search that recalls strong repos the first query missed (e.g. 39k★ frameworks). (4) A hybrid ranker fuses five weighted signals into one fit score (below). (5) Leiden community detection (Louvain fallback) groups the merged pool; a relevance-gated rerank places the best repo per community first. (6) GraphRAG answers questions with evidence grounded in graph facts, with uncertainty flags and human feedback.




## Community detection

Leiden runs on the same graph the system already builds — no training, no extra data; on this pool Leiden and the networkx-Louvain fallback return the identical split (ARI = 1.0), so the structure is algorithm-robust. On a query’s pool the repos split into meaningful sub-communities (core RAG / doc-QA chatbots vs. enterprise & assistant apps). A naive round-robin rerank over-diversified — it once promoted a 5★ repo above two major frameworks — so the final rerank is relevance-gated: it diversifies only the top of the list.



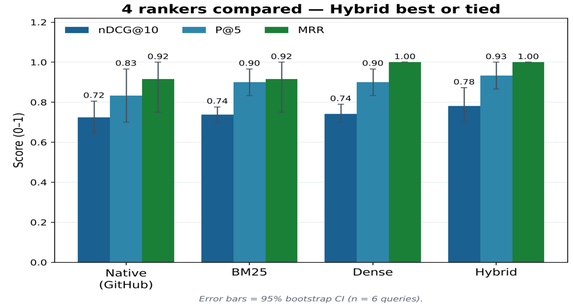

*Real Leiden communities on the live RAG pool (2 communities · 15 repos · 44 edges); Louvain gives the identical split (ARI = 1.0).*

**Why it matters.** The two communities are genuinely different repo families — core RAG / document-QA chatbots vs. enterprise (AWS) and domain assistants. Placing one strong repo per community at the top is exactly what lifts top-5 diversity from 1.7 to 2.3 with no drop in relevance, so the shortlist spans real alternatives instead of near-duplicates.

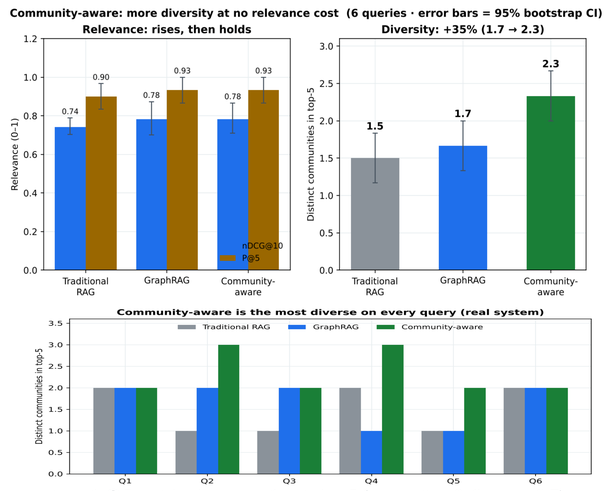
## Results

**Benchmark:** 6 project-idea queries · 90 candidate repos hand-labeled 0 / 1 / 2. Metrics: nDCG@10, P@5, MRR (relevance) and distinct-communities-in-top-5 (diversity). All results carry 95% bootstrap CIs (n=6).

**Hybrid ranker — best or tied on every metric.** nDCG@10 0.78, P@5 0.93, MRR 1.00. At n=6 the nDCG margins over BM25 / Dense fall within the 95% CIs (error bars), so this is a consistent lead, not yet significant.



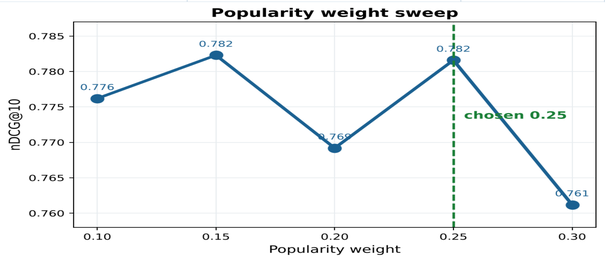

**Community-aware retrieval — the robust win.** In the 3-way ablation it raises diversity +35% (1.7→2.3; paired gain +0.67, 95% CI [+0.17, +1.33], excludes zero) at no relevance cost (ΔnDCG +0.002, CI [−0.01, +0.01]). Every query that differs favours it — 4 / 6 higher, 2 tied, 0 worse.




*diversity@5 = distinct communities in top-5 (pool ceiling 2–3, mean 2.7). Community-aware reaches 2.33 — 87% of achievable — vs 56% for Traditional RAG; the gain holds on every query.*

## What we tried (trial & error)

| Approach tried | Outcome | Decision |
|---|---|---|
| 4 rankers: Native · BM25 · Dense · Hybrid | Hybrid best (nDCG 0.78) | Adopted |
| Popularity weight sweep 0.10 → 0.30 | 0.25 holds relevance + un-buries big repos; 0.30 hurts | 0.25 |
| Untrained GNN (GCN) embedding enrichment | Net-negative; over-smoothing fixed by top-5 cap, still no gain | Rejected |
| Community detection: Leiden vs Louvain vs GNN-clustering | Leiden ≡ Louvain here (ARI 1.0); fast, no training | Leiden |
| Community rerank: pure round-robin | Over-promoted a weak 5★ repo on the live pool | Gated fix |




*Weight sweep: 0.25 holds nDCG and un-buries big repos; 0.30 degrades it.*

## Conclusion

Graph structure + community detection turn repository search from a star-sorted list into a diverse, explainable, trustworthy shortlist: the hybrid ranker handles relevance, communities add diversity, and GraphRAG + human feedback add evidence and control. The honest finding — communities improve diversity, not relevance — is reported plainly, which is the point of evidence-driven Human–AI co-design.

## Future work

Independent ecosystem edges (forks · dependencies · shared contributors) for true ecosystem communities; a larger benchmark via LLM-assisted labeling; a trained GAT on the richer graph; a GraphRAG Q&A + evidence-path panel in the live app.

## References & acknowledgements

Built on: Robertson & Zaragoza, “The Probabilistic Relevance Framework: BM25 and Beyond”; Nussbaum et al., “Nomic Embed: Training a Reproducible Long-Context Text Embedder”.

Stack: Neo4j · Ollama (nomic-embed-text · qwen) · Streamlit · networkx · igraph + leidenalg.